

Shourya Kapoor

B.Tech — Electronics & Computer Engineering — Final Year — Thapar Institute of Engineering and Technology, Patiala
skapoor3_be23@thapar.edu — +91 98555 37077 — github.com/skapoor3be23 — linkedin.com/in/shourya-kapoor-371871348

Objective

Final-year Electronics & Computer Engineering student with hands-on experience building production-style AI applications and intelligent workflows using Large Language Models (LLMs), AI agents, and Retrieval-Augmented Generation (RAG). Skilled in designing REST APIs, optimizing prompts, and working with vector databases for knowledge retrieval. Seeking an Artificial Intelligence Internship to contribute to scalable LLM-powered applications, chatbots, automation systems, and real-world AI products.

Education

B.Tech — Electronics & Computer Engineering 2022 – 2026
Thapar Institute of Engineering and Technology, Patiala — CGPA: 7.64 / 10

Technical Skills

Languages: Python C , C++, object-oriented programming fundamentals
LLMs & AI: Large Language Models (Gemini API, OpenAI-compatible workflows), LangChain, AI Agents, Prompt Engineering, RAG Pipelines, Hallucination Prevention, Chatbots & Virtual Assistants
Retrieval & Data: Vector Databases (FAISS), Embeddings, Document Chunking, Semantic Search, Knowledge Retrieval Systems
Backend & APIs: FastAPI, REST API Design & Integration, JSON, Streamlit, Session Management
Graphics & Interaction: Three.js, WebGL, AR.js, MediaPipe (hand tracking)
Tools & Platforms: Git, GitHub, VS Code, PowerShell, Google AI Studio, Netlify

Projects

AI Research Assistant — Multi-Source RAG with Intelligent Routing June 2026
Python LangChain Gemini API (LLM) FAISS (Vector Database) FastAPI (REST API) Streamlit
Personal Project — github.com/skapoor3be23/ai-research-assistant — Live Demo

Built a multi-source RAG agent and AI automation tool that classifies each query and intelligently routes it to document retrieval, live web search, or general LLM knowledge, rather than relying on a single static source.
Implemented a full Retrieval-Augmented Generation pipeline: PDF ingestion, recursive text chunking, Gemini embeddings, and FAISS vector database indexing with support for multiple documents uploaded at runtime.
Engineered a two-layer hallucination prevention system combining a similarity-score relevance threshold with an LLM self-check, so the agent declines to answer rather than guessing when retrieved content is insufficient.
Added query rewriting to resolve pronouns in follow-up questions (e.g. "its maximum demand") into standalone queries, fixing a retrieval failure discovered through manual testing.
Built conversation memory with automatic history compression to prevent context-window overflow in long sessions.
Exposed the full pipeline as a FastAPI REST API endpoint (POST /ask) returning structured JSON with the answer, source used, and citations; also built a Streamlit chatbot-style chat interface with multi-document upload.

3D Geometric Transformation Visualizer — AR/VR Learning Environment April 2026
Three.js MediaPipe A-Frame AR.js JavaScript
Academic Team Project (3 members) — github.com/Saksham-Mist/AR-VR-Project-Team Rankers

Collaborated in a 3-person team to build an interactive AR/VR platform that visualizes 3D geometric transformations for teaching linear algebra, supporting 5 Platonic solids across desktop, hand-gesture, and marker-based AR interaction modes.
Designed the linear algebra feature set, including real-time $T \times R \times S$ matrix decomposition and step-by-step transformation sequence visualizations demonstrating why matrix multiplication order changes outcomes.
Built the desktop UI/UX layer in Three.js: a synchronized 4-viewport system (orthographic + perspective views) with live matrix display and interactive transformation controls.
Deployed as a publicly accessible live application via Netlify, running at 60 FPS on desktop and 45–55 FPS on mobile.

Courses & Certifications

GenAI Powered Data Analytics Job Simulation June 2026
Tata iQ, via Forage

Completed a self-paced simulation of an AI transformation consultant role, covering predictive modeling with GenAI tools, data exploration and quality assessment, and designing AI-driven systems for fair and responsible automation.

Additional

Actively building production-style AI applications and intelligent software products independently — researching emerging AI technologies and debugging real environment and API-integration issues end to end, from local setup to a deployed, testable system.

Member, Helix Society, Thapar Institute of Engineering and Technology — the institute's annual technical festival.